CTIP — Clinical Trial Intelligence Platform

Parking Lot Item: Sponsor Pipeline Scraper & URL Resolver

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1. Problem Statement

The AACT intervention name field is frequently incomplete, generic, or incorrect. Specific failure modes already documented in the trial_getter_3 specification include:

- Sponsor-proprietary compound codes with no public registry entry (e.g. ABBV-, IASO-, QL-, GR- prefixes)
- Regimen codes without component names (e.g. CarCyDex, Dara-VRD)
- Generic placeholders (e.g. ‘CAR-T cells’ without target antigen)
- Diagnostic assays appearing as interventions (e.g. ELISA)

For trials where multi-field extraction (intervention name + brief title + brief summary) and ChEMBL/DrugBank lookup both fail to resolve a confident drug name, a further enrichment step is warranted. The most authoritative fallback source for a sponsor’s own compound names is the sponsor’s public pipeline page.

2. Proposed Solution Overview

A two-component module — the Sponsor URL Resolver and the Pipeline Page Scraper — operating as a deterministic enrichment step within Phase 4, triggered only for trials flagged as unresolved after all upstream cleaning and lookup steps have been exhausted.

This module is deliberately not a classification step. It produces a drug name candidate and a confidence score, writing the result back to the compounds table with a source tag of sponsor_pipeline. The selected extraction model (to be determined via LM Studio evaluation — see Section 6) then receives this enriched record at the classification stage alongside all other context sources.

3. Component Architecture

3.1 Sponsor URL Resolver

Resolves a sponsor name string (as it appears in AACT) to the URL of the sponsor’s public pipeline page. Operates as a layered funnel — each layer is attempted in order, stopping at the first successful resolution.

Layer	Method	Coverage	Notes
1	Curated lookup table	~40–50% of flagged trials	Static JSON mapping major sponsor names to known pipeline URLs. Highest confidence, zero failure risk. Maintained manually — add entries as new sponsors are encountered.
2a	PMID affiliation & email harvesting	High — pharma-specific	Byproduct of Phase 4b PubMed retrieval already running per compound. Harvest corresponding author email domains and affiliation URL strings from returned abstracts. Email suffix directly resolves the sponsor domain (e.g. @abbvie.com → abbvie.com). Affiliation strings frequently contain explicit URLs. No additional API calls required — data is already in-pipeline. Likely highest-confidence automated resolution after Layer 1 for pharma sponsors specifically, because affiliation is tied to researchers working on the compound rather than a name-string match.
2b	Clearbit Name-to-Domain API	Mid-size western biopharma	Free tier: 50k requests/month. Returns domain from company name. Used when Layer 2a yields no email or affiliation URL. Append common pipeline path patterns and probe which resolves (e.g. /science/pipeline, /research/pipeline, /pipeline).
3	Constrained web search	Smaller sponsors	Query: "{sponsor_name} drug pipeline". Filter results to sponsor's own domain. First matching result taken as candidate URL.
4	LLM fallback (model TBD — see Section 6)	Residual / Chinese biotechs	Pass sponsor name + trial indication to the selected model. Model reasons about alternative sources: SEC filings, press releases, conference abstracts. Returns enrichment from existing corpus rather than pipeline page scrape. Model selection to be confirmed via LM Studio evaluation before build.

Sponsor Name Normalisation (Pre-Resolution)

AACT sponsor name strings are inconsistent across trials. Before any resolution attempt, the sponsor string must be normalised:

- Strip legal suffixes using cleanco (LLC, Inc, GmbH, Co. Ltd., Pharmaceuticals, Biosciences, Therapeutics, etc.)

- Collapse whitespace and punctuation variants
- Apply a known-alias lookup for major sponsors with multiple AACT name variants (e.g. Janssen / Janssen Research & Development / Janssen Pharmaceuticals → janssen)

Layer 2a — PMID Affiliation & Email Harvesting (Detail)

PubMed abstract records returned during Phase 4b compound enrichment carry metadata fields that are a rich, already-in-pipeline source of sponsor domain information. No additional API calls are required — this data arrives as a byproduct of retrieval already running.

Fields to harvest per returned abstract:

- Corresponding author email — the domain suffix directly resolves the sponsor URL (e.g. corresponding.author@abbvie.com → abbvie.com). This is the highest-confidence signal: the email domain is unambiguous and requires no fuzzy matching.
- Affiliation strings — author affiliation fields frequently contain explicit institutional URLs or at minimum a parseable organisation name and location. Example: 'AbbVie Inc., North Chicago, IL. Electronic address: research@abbvie.com' yields the domain directly from the email, or abbvie.com can be inferred from the company name at higher confidence than a generic name-to-domain lookup because the affiliation is tied to the compound being queried.
- Funding and grant fields — PMC full-text records sometimes name the sponsoring organisation in structured funding acknowledgement fields, providing a secondary confirmation signal.
- Full-text links — PMC records may link to the sponsor's own publication page, sharing a domain with the pipeline page.

Implementation note: harvest affiliation and email fields during the existing Phase 4b PubMed retrieval pass and store them in the compounds table alongside the pubmed_context block. The Sponsor URL Resolver then reads these fields before attempting Clearbit or web search. This keeps all PubMed interaction in a single pipeline stage and avoids a second retrieval pass.

Accuracy advantage over Clearbit: affiliation harvesting is tied to the specific compound being queried, not a sponsor name string match. A sponsor with an ambiguous or common name (e.g. 'Synthesis Medical') that might return a wrong domain from Clearbit will yield the correct domain from the email of the researcher who published the compound's Phase I data.

3.2 Pipeline Page Scraper

Given a resolved pipeline URL, extracts the drug name corresponding to the flagged trial. Operates in two stages: page retrieval and entity extraction.

Page Retrieval

- Primary tool: ScrapeGraph-AI (open source, MIT licence, GitHub: ScrapeGraphAI/Scrapegraph-ai)
  - Uses Playwright under the hood — handles JavaScript-heavy pages, lazy loading, and dynamic content
  - Supports Ollama as LLM backend — compatible with all locally-running candidate models; model selection to be confirmed via LM Studio evaluation (Section 6)
  - Accepts a user prompt and source URL; returns structured extraction
- Fallback: Crawl4AI (open source, Apache 2.0, GitHub: unclecode/crawl4ai)
  - Converts pipeline page to clean LLM-ready Markdown
  - Used when ScrapeGraph-AI fails due to page structure complexity

Entity Extraction Prompt

ScrapeGraph-AI is given the following prompt against the resolved pipeline URL:

```
"Extract all pipeline programs as a JSON array. For each program
include: drug_name (INN or internal code), indication,
development_phase, and mechanism_of_action if present. Return only
JSON, no preamble."
```

Trial Matching

The extracted pipeline array is matched against the flagged trial using indication string + phase as the join key. Match confidence is scored on:

- Indication fuzzy match score (rapidfuzz) against trial condition field
- Phase concordance (pipeline phase vs AACT phase)
- Sponsor concordance (already established by the URL resolver)

The highest-scoring match above a configurable threshold (suggested: 0.75) is written back to the compounds table. Matches below threshold go to the human review queue.

4. Data Flow

Step	Input	Output	Owner
Flag unresolved trials	Compounds table post-cleaning	List of NCT IDs with null/low-confidence intervention names	Phase 2 detection tool (existing)
Normalise sponsor name	AACT sponsor field	Cleaned sponsor string	Sponsor URL Resolver

Step	Input	Output	Owner
Harvest PMID metadata (Layer 2a)	PubMed abstracts already retrieved in Phase 4b	Corresponding author email domain + affiliation URL strings per compound	Phase 4b PubMed retrieval (extended)
Resolve pipeline URL	Cleaned sponsor string + PMID-harvested domain (if available)	Pipeline page URL + resolution layer used	Sponsor URL Resolver
Fetch pipeline page	Pipeline URL	Structured pipeline program list (JSON)	Pipeline Page Scraper
Match to trial	Pipeline JSON + trial NCT ID	Drug name candidate + confidence score	Pipeline Page Scraper
Write back	Drug name candidate	compounds table updated with source = sponsor_pipeline	Pipeline Page Scraper
Classification	Enriched record including sponsor_pipeline candidate	Mechanism label (as normal)	LLM B — model TBD via Section 6 evaluation

5. Model Selection — Open Question

Model selection for the extraction and matching step is explicitly deferred pending LM Studio evaluation (see Section 6). Unlike the mechanism classification and compound cleaning tasks elsewhere in CTIP, this task does not require pharmacological knowledge. The model selection logic that applied to TxGemma 27B in those contexts does not automatically carry over here.

5.1 Why This Task Is Different

The sponsor pipeline scraping task is fundamentally an information layout and extraction problem. By the time the model is invoked, ScrapeGraph-AI has already rendered the page and returned structured or semi-structured text. The model is asked to:

- Read a table or list of pipeline programs in plain language
- Identify the row whose indication and phase match the flagged trial
- Extract the drug name string from that row
- Return a confidence score reflecting match quality

No pharmacological reasoning is required. The drug name is already present in plain text on the page — it simply needs to be found and returned. This is a materially different capability profile from mechanism classification or compound name resolution.

5.2 Candidate Models

Model selection is an open question. All four candidates are available locally and should be evaluated using the LM Studio protocol in Section 6 before any code is written.

Model	Strengths for This Task	Size / Speed / Context Window
OpenScholar 8B	Retrieval-augmented by design; well-suited to structured text extraction tasks; strong at identifying relevant content within a larger page context	8B parameters — fast; lightweight inference; low memory footprint. Context window: 128K tokens (fine-tuned on Llama 3.1 8B). Well-suited for large sponsor pipeline pages with many programs.
Mistral 7B	Good structured output reliability on well-defined extraction tasks; clean JSON compliance; straightforward instruction following	7B parameters — fastest of the four candidates; lowest resource cost. Context window: 32K tokens. Sufficient for most pipeline pages; may truncate very large sponsor portfolios.
LLaMA 3 8B	Strong general reasoner; reliable instruction following; good confidence expression	8B parameters — fast; broadly available via Ollama and LM Studio. Context window: 8K tokens (base model). Tight for this task — a large pipeline page plus system prompt could approach the limit.
TxGemma 27B	Excellent structured output reliability and confidence calibration documented in prior pipeline benchmarks; already running locally	27B parameters — 3–4x slower than the 7–8B candidates; highest memory requirement. Context window: 8K tokens (based on Gemma 2). Most constrained of the four — a sponsor with a large oncology pipeline could strain this limit after page extraction and prompt overhead are accounted for.

Note on prior benchmark results: TxGemma 27B outperformed Mistral 7B in previous CTIP evaluations, but those benchmarks tested pharmacological reasoning and compound name resolution — tasks that required biomedical knowledge. Those results do not transfer to an information layout extraction task. All four models should be treated as equal candidates here until the Section 6 evaluation is run.

Note on context window: this task feeds a rendered pipeline page (potentially hundreds of programs for large sponsors) plus a system prompt and trial details into the model. TxGemma 27B and LLaMA 3 8B both have 8K token context windows — the tightest of the group. For a major sponsor like Roche or Novartis with extensive oncology pipelines, this could be a binding constraint. OpenScholar 8B (128K) and Mistral 7B (32K) have a practical advantage here independent of any quality considerations. This should be verified empirically during the LM Studio evaluation by measuring how much context the synthetic test page actually consumes.

Working hypothesis: OpenScholar 8B is the strongest candidate on paper — retrieval-augmented design plus the largest context window. Mistral 7B is the practical fallback if speed is the priority. TxGemma 27B faces a real context window constraint for this specific task that was not a factor in its prior benchmark wins. See Section 6 for the decision rule.

6. LM Studio Evaluation — Suggested Protocol

Following the project's established principle of evaluating before committing, the model selection for this task should be resolved with a focused LM Studio probe before any pipeline code is written. The evaluation is deliberately narrow — it tests information layout extraction, not biomedical reasoning.

6.1 Synthetic Test Page (Contextual Data)

Rather than relying on a live sponsor website (which may change or be unavailable), construct a synthetic pipeline page excerpt representative of what ScrapeGraph-AI would return after rendering a real pharma pipeline page. This gives a controlled, reproducible test input.

Paste the following block as the context in each LM Studio session before presenting the prompt:

```
--- SYNTHETIC SPONSOR PIPELINE PAGE (AbbVie Inc.) ---
Pipeline Programs | Oncology & Haematology

Program         | Indication                          | Phase      | Mechanism
---------------|-----------------------------------|-----------|----------
----------
ABBV-CLS-484    | Solid tumours (PD-1 refractory)    | Phase I    | CD47
antagonist
Navitoclax      | Myelofibrosis                      | Phase II   | BCL-
2/BCL-XL inhibitor
ABBV-383        | Relapsed/Refractory Multiple       | Phase I    | BCMA x
CD3 bispecific
                | Myeloma                            |            |
Venetoclax      | AML (newly diagnosed, fit)         | Phase III  | BCL-2
inhibitor
ABBV-781        | Non-small cell lung cancer         | Phase I    | TIGIT
antagonist
Telisotuzumab   | NSCLC (c-Met overexpressing)       | Phase II   | c-Met
ADC
vedotin         |                                    |            |

--- END OF PAGE CONTENT ---
```

This synthetic page is designed to test four distinct extraction challenges in a single context block:

- A clean proprietary code match (ABBV-383 → BCMA x CD3 bispecific, relapsed myeloma Phase I)
- A clean INN match (Navitoclax → myelofibrosis Phase II)
- A multi-line / wrapped cell requiring the model to join split rows (ABBV-383 indication spans two rows; Telisotuzumab vedotin name is split)
- A case where indication is ambiguous and phase must disambiguate (Venetoclax appears in multiple oncology trials — AML Phase III here)

6.2 Test Prompts

Present each prompt independently in a fresh LM Studio session with the synthetic page above as context. Use identical wording across all candidate models. Record the raw output verbatim.

Prompt A — Clean Proprietary Code Match

```
You are a data extraction assistant. A pharmaceutical company pipeline
page has been retrieved and its content is provided above. A clinical
trial record has an unresolved intervention name. Your task is to identify
the matching pipeline program and return the drug name.  Trial details:
Sponsor: AbbVie Inc.   Condition: Relapsed/Refractory Multiple Myeloma
Phase: Phase I   Intervention name in AACT: [MISSING]  Return a JSON
object with these fields only:   drug_name: the program name or code as it
appears on the pipeline page   match_confidence: a float between 0.0 and
1.0   match_basis: a brief explanation of why you matched this program
Return only JSON. No preamble.
```

Prompt B — Split / Wrapped Cell

```
You are a data extraction assistant. A pharmaceutical company pipeline
page has been retrieved and its content is provided above. A clinical
trial record has an unresolved intervention name. Your task is to identify
the matching pipeline program and return the drug name.  Trial details:
Sponsor: AbbVie Inc.   Condition: Non-small cell lung cancer, c-Met
overexpressing   Phase: Phase II   Intervention name in AACT:
Telisotuzumab  The intervention name in the trial record may be
incomplete. Return the full drug name as it appears on the pipeline page,
including any suffix or qualifier.  Return a JSON object with these fields
only:   drug_name: the full program name as it appears on the pipeline
page   match_confidence: a float between 0.0 and 1.0   match_basis: a
brief explanation of why you matched this program  Return only JSON. No
preamble.
```

Prompt C — Phase-Disambiguated Match

```
You are a data extraction assistant. A pharmaceutical company pipeline
page has been retrieved and its content is provided above. A clinical
trial record has an unresolved intervention name. Your task is to identify
the matching pipeline program and return the drug name.  Trial details:
Sponsor: AbbVie Inc.   Condition: Acute Myeloid Leukaemia, newly diagnosed
```

```
Phase: Phase III   Intervention name in AACT: BCL-2 inhibitor combination
Return a JSON object with these fields only:   drug_name: the program name
as it appears on the pipeline page   match_confidence: a float between 0.0
and 1.0   match_basis: a brief explanation of why you matched this program
Return only JSON. No preamble.
```

Prompt D — No Match / Correct Abstention

```
You are a data extraction assistant. A pharmaceutical company pipeline
page has been retrieved and its content is provided above. A clinical
trial record has an unresolved intervention name. Your task is to identify
the matching pipeline program and return the drug name.  Trial details:
Sponsor: AbbVie Inc.   Condition: Diffuse Large B-Cell Lymphoma   Phase:
Phase II   Intervention name in AACT: [MISSING]  If no program on the
pipeline page matches this trial, return a match_confidence of 0.0 and
explain why in match_basis.  Return a JSON object with these fields only:
drug_name: the matched program name, or null if no match
match_confidence: a float between 0.0 and 1.0   match_basis: explanation
Return only JSON. No preamble.
```

6.3 Scoring Criteria

Score each model on each prompt. Record raw output before scoring.

Criterion	What to Look For	Weight
Correct extraction	drug_name matches expected value exactly or with acceptable normalisation (e.g. capitalisation differences acceptable; missing suffix is a fail)	High
JSON compliance	Returns valid JSON only — no preamble, no markdown fences, no explanation outside the object	High
Confidence calibration	match_confidence is plausible — high for clean matches, low for weak or absent matches. Prompt D should return 0.0 or near-zero.	High
Correct abstention (Prompt D)	Returns null drug_name and low confidence when no match exists. Hallucinating a drug name for DLBCL when none appears on the page is a disqualifying failure.	Critical
Split cell handling (Prompt B)	Correctly joins ‘Telisotuzumab vedotin’ across the wrapped row rather than returning only ‘Telisotuzumab’	Medium
Latency	Record wall-clock time per response. A 3–4x speed penalty (as documented for TxGemma vs Mistral) is acceptable only if output quality is materially better.	Low

6.4 Expected Outcomes & Decision Rule

If OpenScholar 8B or Mistral 7B produce correct extraction, valid JSON, and appropriate abstention on all four prompts — select the faster model. There is no justification for the overhead of TxGemma 27B if a smaller model performs equivalently on a layout task. Note also that TxGemma 27B's 8K token context window is a practical constraint for large sponsor pipelines — if the synthetic test page plus prompt overhead is already consuming a significant fraction of that budget, this should factor into the selection decision even if output quality is comparable.

If any model fails Prompt D (hallucinates a drug name when no match exists) — disqualify it for this task regardless of performance on Prompts A–C. Silent false positives are more damaging to the pipeline than low recall, since a hallucinated drug name would propagate into LLM B classification with a sponsor_pipeline source tag.

If all small models fail on Prompt B (split cell handling) but TxGemma 27B succeeds — accept the latency penalty and use TxGemma 27B. Document the specific failure mode so it can be addressed with prompt engineering in a future iteration.

7. Key Design Constraints

6. Key Design Constraints

- **PMID metadata harvesting is in-pipeline — affiliation and email domain extraction extends the existing Phase 4b PubMed retrieval pass. No second retrieval call, no new external dependency. Fields are stored in the compounds table alongside existing pubmed_context.**
- **Fallback only — this module triggers exclusively for trials that remain unresolved after all upstream steps (multi-field extraction, NER, ChEMBL, DrugBank). It is not a replacement for existing enrichment.**
- **Deterministic before AI — URL resolution (layers 1–3) and page fetching are deterministic. The extraction model only enters at the matching step, and is selected via LM Studio evaluation (Section 6).**
- **Lower confidence weight — sponsor_pipeline results are written with a lower base confidence than ChEMBL-resolved names. They inform LLM B but do not override structured lookup results.**
- **Source tagging — all results tagged source = sponsor_pipeline for full auditability, consistent with existing [ChEMBL], [PubMed], [ASH] tagging conventions.**
- **Human review for weak matches — matches below the confidence threshold (default 0.75) are routed to the existing human review queue, not silently accepted.**
- **Coverage is uneven by design — large western sponsors have clean, scrapeable pipeline pages; Chinese biotechs and very small sponsors may not. The LLM fallback (Layer 4) handles the residual gracefully by routing back to existing corpus enrichment.**

8. Open Source Dependencies

Package	Purpose	Licence	Source
ScrapeGraph-AI	LLM-guided pipeline page extraction	MIT	github.com/ScrapeGraphAI/Scrapegraph-ai
Crawl4AI	Page-to-Markdown conversion (fallback fetcher)	Apache 2.0	github.com/unclecode/crawl4ai
Playwright	Headless browser rendering (via ScrapeGraph-AI)	Apache 2.0	playwright.dev
cleanco	Sponsor name legal suffix stripping	MIT	github.com/psolin/cleanco
Clearbit Name-to-Domain	Company name to domain resolution	Free tier	clearbit.com (50k req/month)
rapidfuzz	Indication fuzzy match scoring	MIT	Already in pipeline

9. Parking Lot Status & Resumption Criteria

This item is deferred to post-Phase 4 stabilisation. It should be revisited once the core enrichment pipeline (ChEMBL, PubMed, conference abstracts) is running cleanly and the residual unresolved compound rate is measured.

Criterion	Detail
Trigger condition	Unresolved compound rate after Phase 4 enrichment exceeds 15% of flagged trials across a full disease run
Prerequisites	Phase 2 detection tool operational; Phase 4 enrichment complete; human review queue instrumented so residual rate is measurable
First build target	Layer 1 (curated lookup table) + Layer 2a (PMID affiliation harvesting) only. Validate coverage before building Layers 2b, 3 and 4. Run LM Studio evaluation (Section 6) in parallel to resolve model selection before any extraction code is written.
Success metric	Reduction in human review queue from current baseline by ≥40% for sponsor-code-specific failures
Owner	To be assigned at Phase 4 retrospective